

GG Diagram: Combined Longitudinal and Lateral Vehicle Acceleration

1 Introduction

The GG diagram provides a representation of the maximum achievable longitudinal and lateral acceleration of a vehicle. It describes the combined acceleration capability of the vehicle in the (a_x, a_y) -plane and is therefore a useful tool for analysing vehicle performance and for determining the acceleration limits subsequently used in lap-time simulation.

The fundamental limitation originates from the available tyre forces. Each tyre can generate a combination of longitudinal and lateral force, but the combined force is limited by the available friction. Consequently, maximum longitudinal acceleration cannot generally be achieved simultaneously with maximum lateral acceleration.

The GG boundary is therefore obtained by determining the maximum feasible acceleration magnitude for every direction in the longitudinal–lateral acceleration plane.

The present model extends the linear bicycle-model force distribution by including longitudinal and lateral load transfer and individual-wheel friction constraints. The vehicle is assumed to be rear-wheel drive. Aerodynamic downforce can additionally be enabled to investigate its influence on the acceleration envelope.

2 Vehicle Model

The vehicle parameters are taken from the linear bicycle model:

$$m = 1500 \text{ kg}, \quad (1)$$

$$I_z = 2500 \text{ kg m}^2, \quad (2)$$

$$a = 1.2 \text{ m}, \quad (3)$$

$$b = 1.4 \text{ m}. \quad (4)$$

The wheelbase is

$$L = a + b = 2.6 \text{ m}. \quad (5)$$

The additional parameters required to calculate load transfer are

$$g = 9.81 \text{ m/s}^2, \quad (6)$$

$$h = 0.5 \text{ m}, \quad (7)$$

$$t_f = 1.6 \text{ m}, \quad (8)$$

$$t_r = 1.6 \text{ m}. \quad (9)$$

Here, h is the height of the centre of gravity above the road surface, while t_f and t_r are the front and rear track widths, respectively.

The parameters h , t_f , and t_r are additional parameters required for the load-transfer calculation and are not contained in the two-degree-of-freedom bicycle model itself.

3 Acceleration Plane

The GG boundary is constructed by considering all possible directions of the acceleration vector. The acceleration magnitude is denoted by R . For an acceleration direction θ ,

$$a_x = R \cos \theta, \quad (10)$$

$$a_y = R \sin \theta, \quad (11)$$

with

$$0 \leq \theta \leq 2\pi. \quad (12)$$

Thus,

$$R = \sqrt{a_x^2 + a_y^2}. \quad (13)$$

For each value of θ , the maximum feasible value

$$R_{\max}(\theta) \quad (14)$$

is determined. The GG boundary is subsequently obtained from

$$a_{x,\text{bnd}}(\theta) = R_{\max}(\theta) \cos \theta, \quad (15)$$

$$a_{y,\text{bnd}}(\theta) = R_{\max}(\theta) \sin \theta. \quad (16)$$

The resulting closed curve represents the maximum combined acceleration capability of the vehicle.

4 Vehicle Force Balance

The required total longitudinal and lateral tyre forces follow directly from the vehicle mass:

$$F_x = ma_x, \quad (17)$$

$$F_y = ma_y. \quad (18)$$

The steady-state bicycle model determines the distribution of lateral force between the front and rear axles.

The lateral force equilibrium is

$$F_{y,f} + F_{y,r} = F_y. \quad (19)$$

The steady-state yaw-moment equilibrium about the centre of gravity is

$$aF_{y,f} - bF_{y,r} = 0. \quad (20)$$

Solving these two equations gives

$$F_{y,f} = \frac{b}{L} F_y \quad (21)$$

and

$$F_{y,r} = \frac{a}{L} F_y. \quad (22)$$

Therefore,

$$F_{y,f} = \frac{b}{L}ma_y, \quad (23)$$

$$F_{y,r} = \frac{a}{L}ma_y. \quad (24)$$

The model assumes rear-wheel drive. Hence,

$$F_{x,f} = 0, \quad (25)$$

$$F_{x,r} = F_x = ma_x. \quad (26)$$

The front axle therefore generates only lateral force, whereas the rear axle generates both longitudinal and lateral force.

5 Left–Right Force Distribution

The total front lateral force is distributed equally between the two front tyres:

$$F_{y,fl} = \frac{F_{y,f}}{2}, \quad (27)$$

$$F_{y,fr} = \frac{F_{y,f}}{2}. \quad (28)$$

Similarly, the rear lateral force is distributed as

$$F_{y,rl} = \frac{F_{y,r}}{2}, \quad (29)$$

$$F_{y,rr} = \frac{F_{y,r}}{2}. \quad (30)$$

The rear longitudinal force is also distributed equally:

$$F_{x,rl} = \frac{F_{x,r}}{2}, \quad (31)$$

$$F_{x,rr} = \frac{F_{x,r}}{2}. \quad (32)$$

Consequently,

$$F_{x,fl} = F_{x,fr} = 0. \quad (33)$$

Although the applied forces are distributed equally, the normal loads of the individual tyres are not necessarily equal because of lateral load transfer.

6 Aerodynamic Downforce

Aerodynamic downforce is modelled using

$$F_{\text{aero}} = \frac{1}{2}\rho C_L A V^2, \quad (34)$$

where

$$\rho = 1.225 \text{ kg/m}^3, \quad (35)$$

$$C_L A = 1.0 \text{ m}^2. \quad (36)$$

The aerodynamic force is distributed between the front and rear axles using

$$\lambda_f = 0.5. \quad (37)$$

Therefore,

$$F_{\text{aero},f} = \lambda_f F_{\text{aero}}, \quad (38)$$

$$F_{\text{aero},r} = (1 - \lambda_f) F_{\text{aero}}. \quad (39)$$

With the present value of λ_f ,

$$F_{\text{aero},f} = \frac{1}{2} F_{\text{aero}}, \quad (40)$$

$$F_{\text{aero},r} = \frac{1}{2} F_{\text{aero}}. \quad (41)$$

When aerodynamic effects are disabled,

$$F_{\text{aero},f} = F_{\text{aero},r} = 0. \quad (42)$$

Since

$$F_{\text{aero}} \propto V^2, \quad (43)$$

the aerodynamic contribution becomes increasingly important with vehicle speed.

7 Longitudinal Load Transfer

During longitudinal acceleration, the inertial effect associated with the centre-of-gravity height produces a longitudinal load transfer between the front and rear axles.

The front axle normal load is

$$F_{z,f} = \frac{mgb - mha_x}{L} + F_{\text{aero},f}. \quad (44)$$

The rear axle normal load is

$$F_{z,r} = \frac{mga + mha_x}{L} + F_{\text{aero},r}. \quad (45)$$

For positive longitudinal acceleration,

$$a_x > 0, \quad (46)$$

the term mha_x/L transfers normal load from the front axle to the rear axle.

For braking,

$$a_x < 0, \quad (47)$$

the load transfer occurs in the opposite direction.

The sum of the axle normal loads is

$$F_{z,f} + F_{z,r} = \frac{mgb - mha_x}{L} + \frac{mga + mha_x}{L} + F_{\text{aero},f} + F_{\text{aero},r} \quad (48)$$

$$= mg + F_{\text{aero}}. \quad (49)$$

Thus, aerodynamic downforce increases the total normal load acting on the tyres.

8 Lateral Load Transfer

During lateral acceleration, the lateral inertial force acting at the centre of gravity produces load transfer between the left and right tyres.

The magnitude of lateral load transfer at the front axle is approximated by

$$\Delta F_{z,f} = \frac{m|a_y|hb}{Lt_f}. \quad (50)$$

At the rear axle,

$$\Delta F_{z,r} = \frac{m|a_y|ha}{Lt_r}. \quad (51)$$

The use of $|a_y|$ means that the magnitude of load transfer is the same for left and right turns. The individual front-wheel normal loads are

$$F_{z,fl} = \frac{F_{z,f}}{2} - \Delta F_{z,f}, \quad (52)$$

$$F_{z,fr} = \frac{F_{z,f}}{2} + \Delta F_{z,f}. \quad (53)$$

The individual rear-wheel normal loads are

$$F_{z,rl} = \frac{F_{z,r}}{2} - \Delta F_{z,r}, \quad (54)$$

$$F_{z,rr} = \frac{F_{z,r}}{2} + \Delta F_{z,r}. \quad (55)$$

The signs simply identify the unloaded and loaded sides. Which physical side corresponds to the minus or plus sign depends on the direction of the turn.

9 Tyre Friction Constraint

The maximum combined tyre force is represented by a friction ellipse.

For a tyre with normal load F_z , the constraint is

$$\left(\frac{F_x}{\mu F_z} \right)^2 + \left(\frac{F_y}{\mu F_z} \right)^2 \leq 1, \quad (56)$$

where μ is the tyre-road friction coefficient.

The equality

$$\left(\frac{F_x}{\mu F_z} \right)^2 + \left(\frac{F_y}{\mu F_z} \right)^2 = 1 \quad (57)$$

represents the limiting condition.

For the front-left tyre,

$$\left(\frac{F_{y,fl}}{\mu F_{z,fl}} \right)^2 \leq 1, \quad (58)$$

because

$$F_{x,fl} = 0. \quad (59)$$

For the front-right tyre,

$$\left(\frac{F_{y,fr}}{\mu F_{z,fr}} \right)^2 \leq 1. \quad (60)$$

At the rear-left tyre,

$$\left(\frac{F_{x,rl}}{\mu F_{z,rl}} \right)^2 + \left(\frac{F_{y,rl}}{\mu F_{z,rl}} \right)^2 \leq 1. \quad (61)$$

At the rear-right tyre,

$$\left(\frac{F_{x,rr}}{\mu F_{z,rr}} \right)^2 + \left(\frac{F_{y,rr}}{\mu F_{z,rr}} \right)^2 \leq 1. \quad (62)$$

Therefore, the candidate acceleration state is feasible only when all four tyres satisfy their respective friction constraints.

10 Normal-Load Feasibility

The friction expressions require positive normal loads. Therefore, the following conditions must also be satisfied:

$$F_{z,fl} > 0, \quad (63)$$

$$F_{z,fr} > 0, \quad (64)$$

$$F_{z,rl} > 0, \quad (65)$$

$$F_{z,rr} > 0. \quad (66)$$

If any wheel reaches

$$F_z \leq 0, \quad (67)$$

the candidate acceleration state is considered infeasible.

This prevents the numerical search from accepting acceleration states for which the load-transfer model predicts an unloaded or lifted wheel.

11 Determination of the GG Boundary

For each acceleration direction θ , the maximum feasible acceleration magnitude R is determined numerically.

The search is initialised using

$$R_{\text{low}} = 0, \quad (68)$$

$$R_{\text{high}} = 30 \text{ m/s}^2. \quad (69)$$

The midpoint of the search interval is

$$R = \frac{R_{\text{low}} + R_{\text{high}}}{2}. \quad (70)$$

The corresponding acceleration components are then calculated from

$$a_x = R \cos \theta, \quad (71)$$

$$a_y = R \sin \theta. \quad (72)$$

All force, load-transfer and friction calculations are subsequently performed for this candidate acceleration.

If all four wheel friction constraints and all positive-normal-load conditions are satisfied, the candidate is feasible and the lower bound is updated:

$$R_{\text{low}} = R. \quad (73)$$

If any constraint is violated, the candidate is infeasible and the upper bound is updated:

$$R_{\text{high}} = R. \quad (74)$$

After repeated iterations, the maximum feasible acceleration magnitude is approximated by

$$R_{\text{max}}(\theta) \approx R_{\text{low}}. \quad (75)$$

The calculation is repeated for all

$$\theta \in [0, 2\pi]. \quad (76)$$

The resulting boundary is

$$a_{x,\text{bnd}} = R_{\text{max}} \cos \theta, \quad (77)$$

$$a_{y,\text{bnd}} = R_{\text{max}} \sin \theta. \quad (78)$$

12 Physical Interpretation of the GG Diagram

The horizontal axis of the GG diagram represents longitudinal acceleration,

$$a_x, \quad (79)$$

while the vertical axis represents lateral acceleration,

$$a_y. \quad (80)$$

The right-hand side of the diagram corresponds to positive longitudinal acceleration, while the left-hand side corresponds to braking.

The upper and lower portions correspond to positive and negative lateral acceleration, respectively.

The boundary therefore describes the maximum combined acceleration available to the vehicle.

At

$$a_y = 0, \quad (81)$$

the vehicle is limited primarily by longitudinal tyre force.

At

$$a_x = 0, \quad (82)$$

the vehicle is limited by lateral tyre force.

For combined braking and cornering, or combined acceleration and cornering, the available longitudinal force is reduced because part of the tyre friction capacity is already being used to generate lateral force.

Consequently, the GG diagram represents the combined rather than independent acceleration limits.

13 Asymmetry Between Acceleration and Braking

The GG diagram is not necessarily symmetric about

$$a_x = 0. \quad (83)$$

For the rear-wheel-drive configuration used here, positive longitudinal acceleration transfers load towards the rear axle:

$$F_{z,r} \uparrow, \quad F_{z,f} \downarrow. \quad (84)$$

This is beneficial for the driven rear tyres because the available rear longitudinal force increases with normal load.

During braking,

$$a_x < 0, \quad (85)$$

normal load is transferred towards the front:

$$F_{z,f} \uparrow, \quad F_{z,r} \downarrow. \quad (86)$$

Since the present model assigns longitudinal force only to the rear axle, the braking-side limitation is strongly affected by the reduction in rear normal load.

Therefore, the longitudinal load-transfer model naturally produces a different acceleration limit during acceleration and braking.

14 Influence of Aerodynamics

The influence of aerodynamic downforce follows directly from

$$F_{\text{aero}} = \frac{1}{2} \rho C_L A V^2. \quad (87)$$

The aerodynamic force increases the normal loads:

$$F_{z,f} = \frac{mgb - mha_x}{L} + F_{\text{aero},f}, \quad (88)$$

$$F_{z,r} = \frac{mga + mha_x}{L} + F_{\text{aero},r}. \quad (89)$$

Since the tyre friction limit is proportional to normal load,

$$F_{\text{tyre,max}} \sim \mu F_z, \quad (90)$$

additional downforce increases the available tyre force.

The effect becomes stronger with increasing vehicle speed because

$$F_{\text{aero}} \propto V^2. \quad (91)$$

For example, with the parameters used in the model,

$$F_{\text{aero}} = \frac{1}{2}(1.225)(1.0)V^2. \quad (92)$$

At $V = 25$ m/s,

$$F_{\text{aero}} = 382.8 \text{ N}. \quad (93)$$

At $V = 50$ m/s,

$$F_{\text{aero}} = 1531.25 \text{ N}. \quad (94)$$

Thus, the difference between the aerodynamic and non-aerodynamic GG boundaries becomes more visible at higher vehicle speeds.

15 Numerical Implementation

The GG calculation is implemented in MATLAB using the function

```
function [ax_bnd, ay_bnd] = ggvdiaqram(V, mu, aero_enabled)
```

The three inputs are:

- V : vehicle speed,
- μ : tyre-road friction coefficient,
- `aero_enabled`: switch controlling aerodynamic downforce.

The acceleration direction is discretised using

```
theta = linspace(0,2*pi,361);
```

giving 361 points over one complete revolution.

For every direction, the maximum feasible acceleration magnitude is determined using 60 binary-search iterations. The resulting acceleration components are calculated as

```
ax_bnd = R_max .* cos(theta);
ay_bnd = R_max .* sin(theta);
```

The final boundary is plotted in the acceleration plane.

16 Model Assumptions

The present GG model uses the following assumptions:

1. The vehicle is represented by the mass and force distribution of the linear bicycle model.
2. The vehicle operates in steady-state lateral equilibrium for the purpose of determining the front/rear lateral-force distribution.
3. The vehicle is rear-wheel drive.

4. Longitudinal force is distributed equally between the two rear tyres.
5. Lateral force is distributed equally between the left and right tyres at each axle.
6. Longitudinal and lateral load transfer are represented using simplified analytical expressions.
7. The tyre force limit is represented by a friction ellipse with a constant friction coefficient μ .
8. Aerodynamic downforce is distributed equally between the front and rear axles.
9. The aerodynamic coefficient C_{LA} is constant and therefore independent of vehicle speed and ride height.
10. The tyre model does not include load sensitivity, combined-slip tyre characteristics, camber effects, relaxation behaviour, or transient tyre dynamics.

These assumptions make the model suitable as a first vehicle-dynamics performance model while retaining the principal coupling between longitudinal acceleration, lateral acceleration, tyre friction and load transfer.

17 Limitations

The friction ellipse used here is a simplified representation of tyre behaviour. In a real tyre, the maximum force is generally not described by a constant- μ ellipse.

In particular, the available tyre force depends on normal load. Therefore, a more detailed model could replace the constant friction coefficient by a load-dependent tyre model.

Furthermore, the present force distribution is obtained from the steady-state bicycle-model equilibrium. It does not explicitly solve the individual tyre slip angles or slip ratios. Consequently, the model does not capture the detailed nonlinear relationship between tyre slip and generated force.

The lateral load-transfer model is also simplified. Suspension geometry, roll stiffness distribution, anti-roll bars, suspension compliance and unsprung masses are not explicitly represented.

Finally, the aerodynamic model uses a constant C_{LA} and a fixed front/rear distribution. A higher-fidelity model would account for the variation of aerodynamic forces with ride height, pitch and vehicle configuration.

18 Connection to Lap-Time Simulation

The GG diagram provides the acceleration limits required by a vehicle-performance model.

For a given vehicle speed and curvature, the required lateral acceleration is

$$a_y = V^2 \kappa, \tag{95}$$

where κ is the path curvature.

The GG boundary determines whether the required lateral acceleration can be achieved and, if so, how much longitudinal acceleration remains available.

Therefore, the GG diagram forms a direct connection between the vehicle-dynamics model and a subsequent lap-time calculation.

A lap-time simulation can use the GG boundary to determine feasible combinations of

$$(a_x, a_y, V)$$

along a given track. This allows the maximum vehicle speed through each corner and the available acceleration during corner exit to be determined.

The resulting longitudinal acceleration capability can subsequently be integrated along the vehicle trajectory to obtain the vehicle speed profile and ultimately the lap time.

19 Summary

The implemented GG model determines the boundary of achievable combined longitudinal and lateral acceleration by combining the bicycle-model force distribution with individual-wheel friction constraints.

The calculation proceeds as follows:

$$(a_x, a_y) \rightarrow (F_x, F_y) \rightarrow (F_{y,f}, F_{y,r}) \rightarrow (F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}) \rightarrow \text{friction constraints} \rightarrow R_{\max}(\theta). \quad (96)$$

The final boundary is

$$\boxed{(a_{x,\text{bnd}}, a_{y,\text{bnd}}) = (R_{\max} \cos \theta, R_{\max} \sin \theta)}. \quad (97)$$

The model therefore captures the principal mechanisms governing the vehicle acceleration envelope: tyre friction, longitudinal load transfer, lateral load transfer, rear-wheel drive and aerodynamic downforce. It also provides the required acceleration-limit representation for subsequent vehicle-dynamics and lap-time-simulation development.